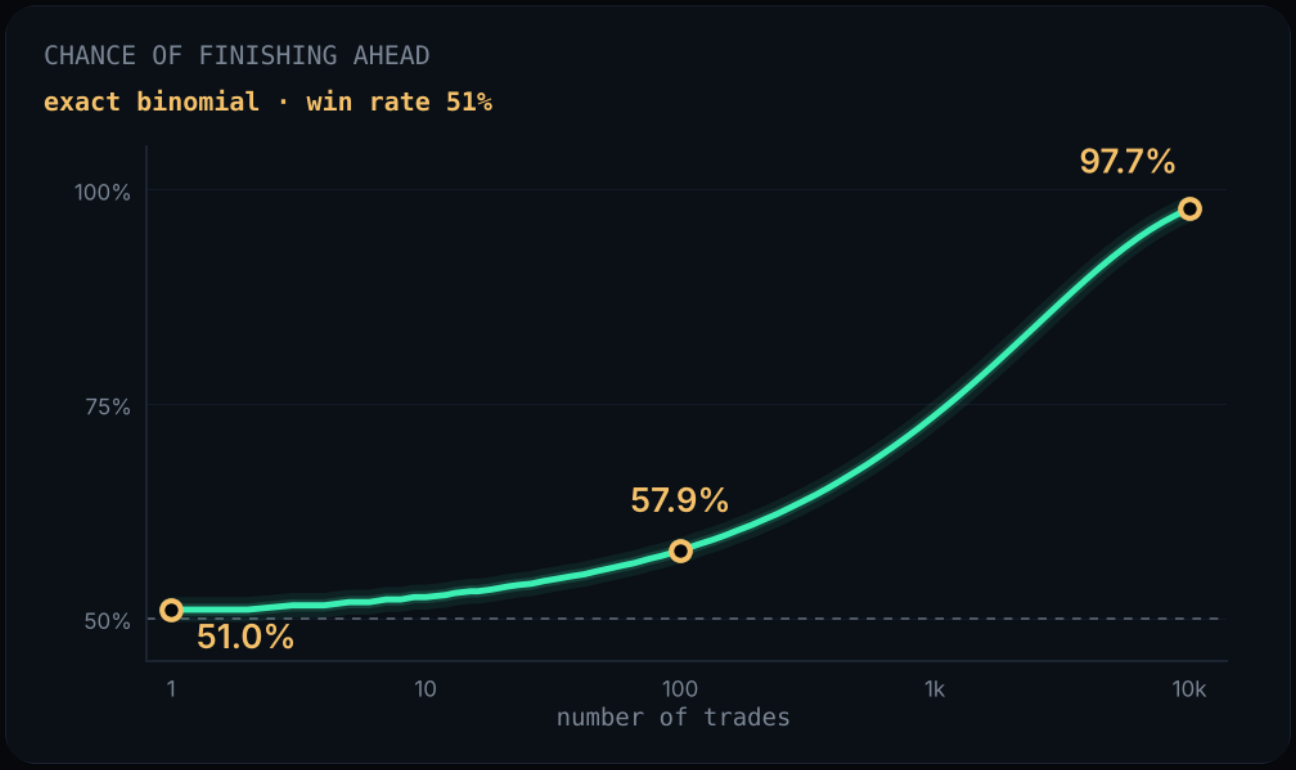
ZERO TO QUANT TRADER

EP. 01 · BEGINNER · FREE GUIDE

What a Quant Does

a tiny edge, repeated



- ✓ all 100 episodes, module by module
- ✓ what to study, in what order
- ✓ tools, books and a 12-week plan

simulated data · educational only · not financial advice · code is for learning, not production trading

1. The model

A strategy makes n independent trades. Each trade pays $+1$ unit with probability p and -1 unit otherwise. Everything in this guide follows from three quantities of one trade.

symbol	meaning	value in the reel
p	win rate, the probability a trade pays $+1$	0.51
$e = 2p - 1$	edge: the average profit of one trade	0.02
$\sigma = \sqrt{1 - e^2}$	standard deviation of one trade	0.9998, about 1
n	number of trades	1, 100, 10,000

Edge. The average of $+1$ (weight p) and -1 (weight $1 - p$):

$$e = p \cdot (+1) + (1 - p) \cdot (-1) = 2p - 1$$

Spread of one trade. The variance is $E[X^2] - e^2 = 1 - e^2$, because $X^2 = 1$ always. So $\sigma = \sqrt{1 - e^2}$.

After n independent trades. Means add, variances add:

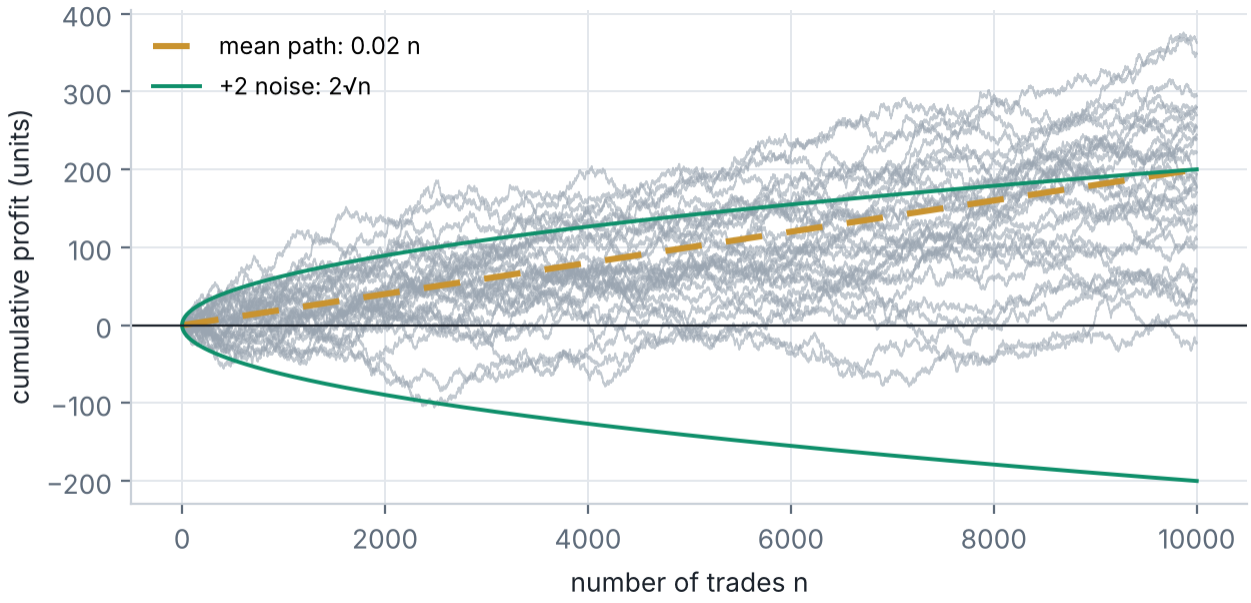
$$\text{mean} = e\,n, \qquad \text{noise} = \sigma\sqrt{n}$$

The mean grows with n, the noise only with $\sqrt{n}$. The ratio is what decides the outcome:

$$z = \frac{e\,n}{\sigma\sqrt{n}} = \frac{e}{\sigma}\sqrt{n}, \qquad P(\text{ahead after } n \text{ trades}) \approx \Phi(z)$$

Φ is the standard normal cumulative distribution: $\Phi(0) = 50\%$, $\Phi(1) = 84\%$, $\Phi(2) = 97.7\%$. The reel's three numbers, computed exactly from the binomial distribution (a tie counts half) and with this formula:

n	mean $e\,n$	noise $\sigma\sqrt{n}$	z	exact	$\Phi(z)$
1	0.02	1	0.02	51.0%	50.8%
100	2	10	0.20	57.9%	57.9%
10,000	200	100	2.00	97.7%	97.7%



30 simulated strategies, p = 0.51. Dashed: mean path. Green: plus and minus two noise widths.

In this simulation 29 of the 30 paths end above zero after 10,000 trades. The median final profit is 177 (the mean is 200). The lowest point reached by any path along the way is −106. The edge does not remove losing stretches. It outlasts them.

2. Three uses

2.1 Is this backtest evidence of an edge?

A backtest shows 200 trades and a 53% win rate. Test it against "no edge": $p = 0.5$, so $e = 0$ and $\sigma = 1$. The observed edge is $\hat{e} = 2(0.53) - 1 = 0.06$, so

$$z = \hat{e}\,\sqrt{n} = 0.06\sqrt{200} = 0.85$$

A strategy with no edge at all produces a win rate of 53% or better in 200 trades 21.8% of the time (exact binomial). That is not evidence. A common threshold is $z \geq 2$. Solving $\hat{e}\sqrt{n} = 2$ for n gives

$$n = \left(\frac{2}{\hat{e}}\right)^2 = \left(\frac{2}{0.06}\right)^2 \approx 1{,}112 \text{ trades}$$

So the same 53% needs about 1,112 trades before it means something. Fewer trades, or a win rate found by trying many variants and keeping the best, proves much less.

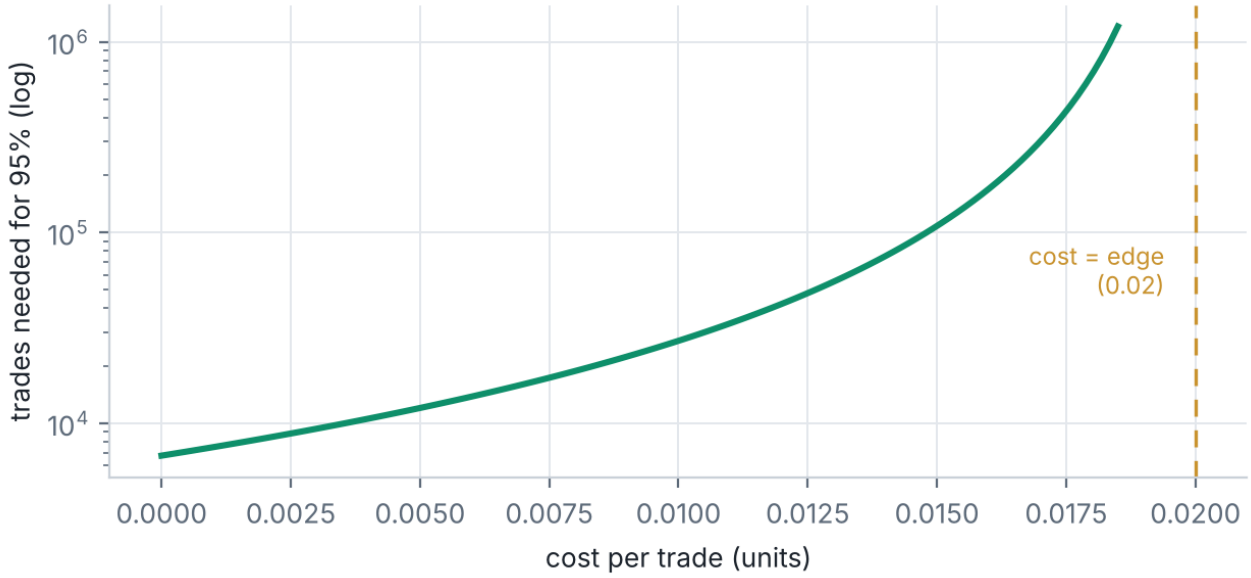
2.2 What costs do to the edge

Each trade pays a cost c (spread, fees, slippage), so the edge that matters is $e_{\text{net}} = 2p - 1 - c$. Trades needed for a 95% chance of finishing ahead, with $z_{95} = 1.645$:

$$n = \left(\frac{z_{95}\,\sigma}{e_{\text{net}}}\right)^2$$

Here $p = 0.51$ ($e = 0.02$):

cost c	net edge	trades needed
0	0.020	6,762
0.005	0.015	12,020
0.010	0.010	27,045
0.015	0.005	108,179
0.020	0	never



Trades needed for 95% against cost per trade, p = 0.51.

A cost of one quarter of the edge nearly doubles the trades needed. A cost equal to the edge makes the strategy a coin flip, however many trades you make.

2.3 Why trade often: time and Sharpe ratio

With N trades per year, the annualised Sharpe ratio of this strategy (profit per unit of risk, one unit risked per trade) is

$$\mathrm{SR} = \frac{e}{\sigma}\sqrt{N}$$

Same edge, $p = 0.51$, 252 trading days:

trades per day	days to 95% confidence	Sharpe ratio
2	3,381 (13.4 years)	0.45
20	338 (1.3 years)	1.42
200	34 (0.1 years)	4.49

Ten times the trades per day shortens the wait by ten and raises the Sharpe ratio by $\sqrt{10} \approx 3.2$. This is why firms with tiny edges trade often. It is also why the assumptions below matter: the table holds only if the trades are independent and the edge survives the extra volume.

Where the model breaks

- **Trades are not independent.** Positions in the same market move together, so 200 correlated trades carry less information than 200 independent ones, and $\sqrt{n}$ is too optimistic.
- **Payoffs are not ± 1.** Real wins and losses differ in size. What counts is the expected profit per trade, not the win rate. A 40% win rate can be profitable if winners are three times the size of losers.
- **The edge decays.** Others find the same pattern. A p estimated on old data is an upper bound on the future one.
- **Sizing.** An edge does not protect against betting too much on one trade. See the Kelly episodes (39 and 40).

3. Task

Do this before the next episode. About 30 minutes, Python with NumPy.

Part A. Predict. A strategy has win rate $p = 0.505$. You will simulate 200 of them, each with 1,000 trades. Using the formulas in section 1, compute: (a) the edge e; (b) the mean and the noise of the final profit; (c) the chance one strategy ends ahead; (d) how many of the 200 you expect above zero. Write the four numbers down before running anything.

Part B. Simulate. Fill in the blanks:

```python
import numpy as np

rng = np.random.default_rng(7)
p, n, k = 0.505, 1_000, 200
wins  = rng.binomial(n, p, size=k)    # wins in each strategy
final = ____                          # final profit: wins pay +1, losses -1
print((final > 0).sum(), final.mean(), final.std())
```

Compare the three printed numbers with your predictions.

Part C. Costs. Repeat with a cost of $c = 0.01$ per trade, subtracting cn from every final profit. Predict first, using $e_{\text{net}} = e - c$: what happens to the number of strategies above zero, and why?

Part D. Quick questions.

1. $p = 0.55$, 100 trades. Chance of finishing ahead?
2. $p = 0.50$, 10,000 trades. Chance of finishing ahead? What does it say about trading more without an edge?
3. $p = 0.52$. Trades needed for a 99% chance of finishing ahead ($z_{99} = 2.326$)?

The answers follow. Check yours before reading them.

Answers

A. $e = 2(0.505) - 1 = 0.010$. Mean $= 0.010 \times 1000 = 10$. Noise $= \sigma\sqrt{n} = 0.99995 \times 31.6 \approx 31.6$. $z = 10/31.6 = 0.32$, so the chance of finishing ahead is 62.4% and the expected number above zero is $0.624 \times 200 \approx 125$.

B. `final = 2*wins - n`. With seed 7: 127 strategies above zero (63.5%), mean final profit 11.3, standard deviation 32.2. Theory said 125, 10 and 31.6. The gaps are sampling noise of 200 strategies: another seed gives different numbers close to the same.

C. With $c = 0.01$ the net edge is $0.010 - 0.010 = 0$, so $z = 0$ and about half of the strategies end ahead (about 100 of 200), with mean final profit near 0. The cost took the whole edge, and 1,000 trades cannot recover it.

D. (1) 84.1%: $e = 0.10$, $z = 0.10 \times 10 = 1$, $\Phi(1) = 84\%$. (2) 50.0%: no edge means $z = 0$ for every n. More trades widen the spread of outcomes and leave the average at zero. (3) $n = (2.326 \times 0.9992/0.04)^2 \approx 3{,}378$ trades.

4. The 100-episode roadmap

One episode per day from 6 November 2026 to 13 February 2027. Each ends with a comment keyword; comment it under that episode to get the material in your DMs.

Module 1 — Market Basics (Ep. 1–10)

Order books, spreads, costs, who is on the other side.

Ep.	Title	Free with the comment keyword
1	What a quant actually does	**ROADMAP**: the 100-episode roadmap and study plan (PDF)
2	The order book in 30 seconds	**BOOK**: order-book cheat sheet (PDF)
3	The spread you pay without noticing	**SPREAD**: trading-cost calculator (Google Sheet)
4	Market vs limit orders	**ORDERS**: order-types one-pager (PDF)
5	Who's on the other side of your trade	**WHO**: adverse selection explained (PDF)
6	Auctions, ticks and halts	**AUCTION**: market-structure guide (PDF)
7	The quant's toolbox	**TOOLBOX**: instruments cheat sheet with ES, NQ and MNQ specs (PDF)
8	Leverage: one micro contract, $60k of Nasdaq	**LEVERAGE**: leverage and margin calculator (Google Sheet)
9	Simple vs log returns	**RETURNS**: returns and compounding notebook
10	Costs: the silent strategy killer	**COSTS**: transaction-cost checklist (PDF)

Module 2 — Probability & Statistics (Ep. 11–20)

Expected value, variance, distributions, interview puzzles.

Ep.	Title	Free with the comment keyword
11	Expected value: the only number that matters	**EDGE**: win rate and payoff calculator (Google Sheet)
12	Volatility drag	**DRAG**: volatility-drag guide with simulation code (PDF)
13	The normal distribution lies about crashes	**TAILS**: fat-tails notebook, normal vs Student-t
14	Bayes: update like a quant	**BAYES**: Bayes cheat sheet and 10 interview puzzles (PDF)
15	Correlation breaks when you need it	**CORR**: rolling-correlation notebook
16	Why the casino always wins	**CASINO**: law of large numbers and trade count (PDF)
17	Is your backtest just luck?	**TSTAT**: significance-test notebook
18	Regression: alpha vs beta	**ALPHA**: alpha and beta regression notebook
19	Test 100 strategies, find 5 fake winners	**LUCK**: multiple-testing guide: Bonferroni, deflated Sharpe (PDF)
20	Monte Carlo: 10,000 alternate histories	**SIM**: equity-curve resampling notebook

Module 3 — Python for Quants (Ep. 21–28)

Python setup, data, pandas, plotting.

Ep.	Title	Free with the comment keyword
21	A quant setup for $0	**SETUP**: setup guide and requirements file (PDF)
22	20 years of prices in 3 lines	**DATA**: price downloader script
23	pandas for traders	**PANDAS**: pandas cheat sheet (PDF)
24	Survivorship bias	**SURVIVOR**: point-in-time universe builder (code)
25	Plot like a hedge fund	**PLOT**: dark-theme plotting template (same look as the reels)
26	Vectorize or die	**FAST**: numpy vectorization notebook
27	Your first backtest in 20 lines	**BACKTEST**: 20-line backtest with notes (code + PDF)
28	Look-ahead bias	**LOOKAHEAD**: 10 backtest bugs checklist (PDF)

Module 4 — Time Series (Ep. 29–38)

Returns, stationarity, autocorrelation, ARMA, GARCH.

Ep.	Title	Free with the comment keyword
29	Can you beat a coin flip?	**RANDOM**: random-walk generator and "spot the real chart" quiz
30	Stationarity	**STATIONARY**: ADF test notebook
31	Does yesterday predict today?	**ACF**: autocorrelation notebook
32	Mean reversion vs momentum	**REGIME**: Hurst exponent code and guide (PDF)
33	ARIMA in 30 seconds	**ARIMA**: ARIMA notebook
34	Volatility clustering	**CLUSTER**: volatility-clustering chart code (PDF)
35	GARCH: forecasting volatility	**GARCH**: GARCH(1,1) notebook
36	Cointegration	**COINT**: Engle-Granger notebook
37	Pairs trading, tested over 20 years	**PAIRS**: full pairs-trading backtest code
38	Kalman filter	**KALMAN**: adaptive hedge-ratio notebook

Module 5 — Portfolio & Risk (Ep. 39–50)

Mean-variance, Kelly, risk measures, position sizing.

Ep.	Title	Free with the comment keyword
39	Kelly criterion: how much to bet	**KELLY**: Kelly sizing calculator and guide (Google Sheet + PDF)
40	Why pros bet half-Kelly	**HALF**: full vs half Kelly drawdown simulation (notebook)
41	Sharpe, Sortino, Calmar	**SHARPE**: performance-metrics calculator (Google Sheet + Python)
42	The drawdown that ends careers	**DRAWDOWN**: drawdown analyzer code
43	VaR and expected shortfall	**VAR**: VaR and expected shortfall notebook
44	Markowitz: diversification maths	**FRONTIER**: efficient-frontier notebook
45	Why optimizers blow up	**OPTIMIZER**: shrinkage and robust weights guide (PDF + code)
46	Risk parity	**PARITY**: risk-parity weights calculator
47	Volatility targeting	**VOLTARGET**: volatility-targeting code
48	A quant book's portfolios, out of sample	**LAZY**: 25 portfolios backtested (PDF)
49	Why All Weather broke in 2022	**WEATHER**: 2022 post-mortem (PDF)
50	Three engines: survived 2008 and 2022	**ENGINES**: full recipe and backtest code, plus the Ep. 1–50 handbook

Module 6 — Strategies & Backtesting (Ep. 51–64)

Momentum, mean reversion, pairs, honest backtesting.

Ep.	Title	Free with the comment keyword
51	Time-series momentum	**TREND**: time-series momentum code and paper summary
52	The Turtle experiment	**TURTLE**: the original Turtle rules in Python
53	Buy last year's best stock	**WINNER**: full backtest with year-by-year picks
54	Momentum 12-6-3 on the Nasdaq-100	**MOMENTUM**: strategy code and robustness grid
55	Dual momentum	**DUAL**: dual-momentum code
56	Sector momentum, since 1927	**SECTOR**: sector-momentum notebook on Fama-French data
57	Betting Against Beta	**BETA**: Betting Against Beta replication code
58	Combining weak strategies	**COMBO**: equal-risk combination code
59	The last 30 minutes	**CLOSE**: intraday-momentum test code and verdict (PDF)
60	The Noise Area strategy	**NOISE**: full strategy rules and code
61	VWAP: the "holy grail"?	**VWAP**: VWAP strategy code and paper notes
62	Overfitting	**OVERFIT**: overfitting checklist (PDF)
63	Walk-forward testing	**WALK**: walk-forward framework code
64	The maths of a prop-firm challenge	**PROP**: expected-value calculator per firm (Google Sheet)

Module 7 — Stochastic Calculus (Ep. 65–70)

Brownian motion, Ito's lemma, the maths under option pricing.

Ep.	Title	Free with the comment keyword
65	Brownian motion	**BROWNIAN**: Brownian-motion simulator (notebook)
66	Bachelier, 70 years early	**BACHELIER**: the history plus Bachelier vs Black-Scholes code (PDF)
67	Geometric Brownian motion	**GBM**: GBM path simulator
68	Itô's lemma	**ITO**: step-by-step Itô derivation (PDF)
69	Monte Carlo pricing	**MONTE**: Monte Carlo option pricer (notebook)
70	Risk-neutral pricing	**NEUTRAL**: risk-neutral pricing explained (PDF)

Module 8 — Options & Derivatives (Ep. 71–82)

Black–Scholes, Greeks, binomial trees, hedging.

Ep.	Title	Free with the comment keyword
71	Options in 30 seconds	**OPTIONS**: payoff-diagram builder
72	Put-call parity	**ARB**: parity arbitrage sheet (Google Sheet)
73	Binomial trees	**TREE**: binomial pricer (notebook)
74	Black-Scholes from scratch	**PRICE**: Black-Scholes calculator and derivation (PDF + code)
75	Delta hedging	**DELTA**: delta-hedging simulator
76	The Greeks	**GREEKS**: Greeks cheat sheet and calculator
77	Gamma: the Greek that moves markets	**GAMMA**: dealer gamma explained (PDF)
78	GameStop 2021: a gamma squeeze	**SQUEEZE**: gamma-squeeze feedback model (PDF)
79	Theta: paid to wait	**THETA**: time-decay chart code
80	The wheel strategy, quantified	**WHEEL**: wheel rules and paper findings (PDF)
81	Futures and the basis	**CARRY**: futures fair-value calculator
82	Exotic options	**EXOTIC**: Monte Carlo pricer for barriers, digitals and Asians

Module 9 — Volatility (Ep. 83–89)

Implied vol, skew, local and stochastic volatility, Heston.

Ep.	Title	Free with the comment keyword
83	Implied vs realised volatility	**VRP**: implied vs realised notebook
84	The volatility smile	**SMILE**: smile and skew guide (PDF)
85	The VIX	**VIX**: how the VIX is calculated, step by step (PDF)
86	Volmageddon, 5 February 2018	**XIV**: post-mortem and VIX futures term structure (PDF)
87	Harvesting the volatility premium	**VOLEDGE**: the paper's rules in code
88	Heston: volatility with a life of its own	**HESTON**: derivation, commented code, semi-analytic pricing and calibration (PDF)
89	The volatility surface	**SURFACE**: volatility-surface builder (notebook)

Module 10 — Machine Learning (Ep. 90–95)

Features, validation, models that survive out-of-sample.

Ep.	Title	Free with the comment keyword
90	ML in trading: what actually works	**BOT**: map of ML methods that hold up in markets (PDF)
91	Feature engineering	**FEATURES**: 30 price and volume features (notebook)
92	Random forests and XGBoost	**XGB**: classifier with walk-forward validation (notebook)
93	LSTMs and transformers: hype check	**LSTM**: LSTM notebook and reality check
94	Reinforcement-learning traders	**RL**: RL trading environment (code)
95	LLMs that invent alpha	**LLM**: prompt pack and factor-testing code

Module 11 — Execution & Going Live (Ep. 96–100)

Transaction costs, execution, running a strategy live.

Ep.	Title	Free with the comment keyword
96	Market impact: the square-root law	**IMPACT**: market-impact calculator (Google Sheet)
97	Optimal execution	**EXECUTE**: Almgren-Chriss notebook
98	How market makers make money	**MM**: market-making simulator
99	Knight Capital: $440M in 45 minutes	**KILL**: go-live risk checklist and kill-switch template (PDF)
100	Zero to Quant Trader: the whole map	**QUANT**: the full handbook: all 100 guides plus the roadmap

5. A 12-week plan, 5 to 6 hours a week

weeks	modules	what to do
1	1 Market Basics	Watch episodes 1 to 10. For each, write one number from the reel and the line of code that produced it.
2–3	2 Probability & Statistics	Redo every expected-value example by hand, then in Python. Do the interview puzzle before watching the answer.
4	3 Python for Quants	Set up the environment, download prices, plot returns in one notebook.
5–6	4 Time Series	Run each test on two different assets and note where the result changes.
7	5 Portfolio & Risk	Build a two-asset portfolio and compute its risk three ways.
8–9	6 Strategies & Backtesting	Backtest one simple rule with costs, then try to break it on a different period (section 2.1 applies).
10	7 and 8	Price a call by simulation and by formula and compare.
11	9 Volatility	Compute realised volatility and compare it with a published implied volatility.
12	10 and 11	Fit one simple model with a train/test split and estimate what execution would cost (section 2.2 applies).

Required background. School algebra and basic probability. Modules 1 and 2 need no programming. Module 3 starts from an empty notebook. Python 3 with NumPy, pandas, SciPy and matplotlib, all free, is enough for almost the whole series.

6. Books

- Hull, J. *Options, Futures, and Other Derivatives* (Pearson).
- Shreve, S. *Stochastic Calculus for Finance II* (Springer, 2004).
- Grinold, R. and Kahn, R. *Active Portfolio Management* (McGraw-Hill, 2nd ed., 1999).
- Carver, R. *Systematic Trading* (Harriman House, 2015).
- Harris, L. *Trading and Exchanges* (Oxford University Press, 2003).
- Jansen, S. *Machine Learning for Algorithmic Trading* (Packt, 2nd ed., 2020).
- López de Prado, M. *Advances in Financial Machine Learning* (Wiley, 2018).

Education only, not financial advice. All numbers in sections 1 to 3 are simulated or computed from the model; the code is in guide_code.py.